Induced Subgraphs in Ramsey Graphs

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Abstract

We study the "random-like" features of Ramsey graphs, focusing on the sizes of their induced subgraphs.

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1 Introduction

Classical Ramsey Theory asks for the minimum number of vertices required to guarantee the existence of a clique or an independent set in a graph. We refer to these maxima as *Ramsey numbers*.

Definition 1.1. The *Ramsey number* $R(s,t)$ is the minimum number N such that any graph on N vertices contains a clique of size s or an independent set of size t.

The existence of these numbers was proven by Ramsey [12] in 1930, but their quantitative behaviors remain largely unknown despite decades of intensive research. In particular, it is known that $R(3,3) = 6$ and $R(4,4) = 18$, but the exact values of $R(5,5)$ and beyond are yet to be determined. The cornerstone of Ramsey Theory due to Erdős and Szekeres in 1935 gives the following upper bound on $R(s,t)$:

Theorem 1.1 (Erdős–Szekeres [6])**.** *For $s,t \geq 2$,*

$$R(s,t) \leq \binom{s+t-2}{s-1}.$$

Following the fact that $\binom{2t-2}{t-1} < 2^{2t}$, we can see $R(t,t) < 4^t$. Very little progress had been made on this bound until the recent breakthrough of Campos, Griffiths, Morris, and Sahasrabudhe [4], giving the first exponential improvement since the result of Erdős and Szekeres. On the other hand, the lower bound of Ramsey numbers has been even harder to determine. A rather basic result which already gives the best known exponents is the famous random graph construction due to Erdős in 1947, marking one of the first applications of the probabilistic method in combinatorics. Before stating the result, we define the *Erdős–Rényi random graph*.

Definition 1.2. For $n \geq 1$ and $p \in [0,1]$, the *random graph* $G_{n,p}$ is the graph on n vertices in which each of the $\binom{n}{2}$ edges is present independently with probability p.

Remark: For constant p, the random graph $G_{n,p}$ is dense in the sense that it has $\Theta(n^2)$ many edges almost surely.

Theorem 1.2 (Erdős [7])**.** *For $t \geq 3$,*

$$\sqrt{2}^t < R(t,t) < 4^t.$$

Proof. The upper bound follows from Theorem 1.1. For lower bound, set $N = \lfloor \sqrt{2}^t \rfloor$ and consider the random graph $G_{N,1/2}$. The expected number of cliques of size t is

$$\binom{N}{t} \cdot 2^{-\binom{t}{2}} < \frac{N^t}{t!} 2^{-\binom{t}{2}} \leq \frac{\sqrt{2}^{t^2}}{t!} 2^{-\binom{t}{2}} = \frac{\sqrt{2}^t}{t!} = \frac{\sqrt{2}^{t-3}}{t(t-1)\cdots 4} \cdot \frac{\sqrt{2}^3}{3!} \leq \frac{\sqrt{2}^3}{3!} < \frac{1}{2}.$$

Similarly, the expected number of independent sets of size t is less than $1/2$. It now follows that there exists a graph on N vertices containing neither a clique nor an independent set of size t. $\square$

We define a shorthand for cliques and independent sets.

Definition 1.3. A set $S \subseteq V(G)$ is *homogeneous* in a graph G if it induces a clique or an independent set in G. We denote the size of the largest homogeneous set in G by $\hom(G)$.

Therefore, Theorem 1.2 may be restated as

$$\frac{1}{2}\log_2 n < \hom(G) < 2\log_2 n,$$

for any graph G on n vertices. Apart from determining the bounds of Ramsey numbers, the search for explicit graph constructions without large homogeneous sets has also been a major open problem in Ramsey Theory. We refer to such graphs as *Ramsey graphs*.

Definition 1.4. Let $C > 0$ be a constant. A *C-Ramsey graph* is a graph with no homogeneous set of size greater than $C \log n$.

Despite decades of intensive research, all known constructions of Ramsey graphs still rely on some randomness. Thus it is believed that Ramsey graphs inherently resemble dense random graphs like $G_{n,1/2}$, and there are a number of theorems showing that certain "richness" properties of random graphs also hold for Ramsey graphs. The first result that cemented this connection is due to Erdős and Szemerédi, showing that Ramsey graphs are also dense.

Theorem 1.3 (Erdős–Szemerédi [7])**.** *Let G be a graph on n vertices with edge density $d \le 1/2$. Then G contains a homogeneous set of size greater than $K = \frac{c}{d\log(1/d)}\log n$, for some constant $c > 0$ and sufficiently large n.*

Proof. Suppose not. Pick $c < 1/32$. Since G contains $d\binom{n}{2}$ edges, there exists $m \ge n/2$ vertices in G with degree at most dn, by the pigeonhole principle. We work with the graph G' induced by these m vertices. Let I be the largest independent set in G' and let $k = |I|$. Since $\deg(v) \le dn$ for all $v \in I$, we have $\sum_{v\in I}\deg(v) \le dnk \le 2dmk$. Put $s = 8dk$. Again by pigeonhole, there are at least $m/2 - k$ vertices $v \in V(G') \setminus I$ with $\deg_I(v) < s$. Let J be the set of these vertices. Consider the family $\mathcal{F}$ of subsets of I with size less than s. Partition J by each vertex's neighbourhood in I, and note that each resulting class corresponds to an element of $\mathcal{F}$. By the standard bounds $\binom{x}{r} \le (ex/r)^r$ and $1 + x \le e^x$,

$$|\mathcal{F}| \le s\binom{k}{s-1} \le s\left(\frac{ek}{s-1}\right)^{s-1} = s\left(\frac{s}{s-1}\right)^{s-1}\left(\frac{ek}{s}\right)^{s-1} \le se\left(\frac{ek}{s}\right)^{s-1} = \frac{s^2}{k}\left(\frac{ek}{s}\right)^{s}.$$

By assumption $|I| = k < K$, so

$$s < 8dK = \frac{8c\log n}{\log(1/d)} = O(\log n).$$

Since $s/k = 8d$,

$$|\mathcal{F}| \le \frac{s^2}{k}\left(\frac{ek}{s}\right)^s \le O(\log n)\cdot\left(\frac{e}{8}\cdot\frac{1}{d}\right)^s \le O(\log n)\cdot\left(\frac{1}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = O(\log n)\cdot n^{8c} < n^{1/3}.$$

But then

$$\frac{|J|}{|\mathcal{F}|} \ge \frac{m/2-k}{n^{1/3}} \ge \sqrt{n},$$

so by pigeonhole, there exists a class of vertices Z in J of size at least $\sqrt{n}$. Let $G'' = G[Z]$. Note that G'' does not contain an independent set of size s, otherwise exchange it with Y in I creates a larger independent set in G', contradicting the maximality of I. Let t be the size of the largest clique in G'', and note that $t < K$ by assumption. Erdős–Szekeres theorem and the standard binomial bounds give

$$\sqrt{n} \le |Z| \le \binom{s+K-2}{s-1} \le \binom{s+K}{s} \le \binom{8dK+K}{8dK} \le \left(\frac{e(8d+1)}{8d}\right)^s \le \left(\frac{2}{d}\right)^s,$$

where the last inequality holds because $e(1+8d)/8 \le 5e/8 \le 2$ for $d \le 1/2$. But then since $d \le 1/2$ and $c < 1/32$,

$$\left(\frac{2}{d}\right)^s \le \left(\frac{2}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = n^{8c\cdot\frac{\log(2/d)}{\log(1/d)}} = n^{8c\left(1+\frac{\log 2}{\log(1/d)}\right)} \le n^{16c} < \sqrt{n},$$

contradiction. □

Recall that the Erdős–Szekeres theorem (Theorem 1.1) gives $\hom(G) \ge \frac{1}{2}\log_2 n$, regardless of the edge density. Theorem 1.3 improves this when the density is small. For example, if $d = 1/n^\varepsilon$ for some $\varepsilon > 0$, then $\hom(G) = \Omega\left(\frac{n^\varepsilon}{\log n^\varepsilon}\log n\right)$, vastly larger than $\frac{1}{2}\log_2 n$. Intuitively, a sparse graph should contain a large independent set, and the theorem makes this notion precise.

If G is C-Ramsey, then applying Theorem 1.3 to whichever G or $\overline{G}$ has density $d \le 1/2$ yields $c/(d\log(1/d)) \le C$, forcing d to be bounded away from 0. By the same argument applied to $\overline{G}$ forces d to be bounded away from 1. This gives the following corollary.

Corollary 1.4. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Due to the rich variety of induced subgraphs in dense random graphs, a popular angle to study the pseudorandom properties of Ramsey graphs is to consider their induced subgraphs. Some notable results are a theorem of Shelah [13] which asserts that every n-vertex Ramsey graph contains $2^{\Omega(n)}$ non-isomorphic induced subgraphs, and a theorem of Prömel and Rödl [11] which shows the existence of $\delta = \delta(C) > 0$ such that every C-Ramsey graph on n vertices contains an induced copy of all graphs on at most $\delta\log n$ vertices.

It is also interesting to look at the edge statistics of induced subgraphs in Ramsey graphs. For a graph G, let

$$\Phi(G) \coloneqq \{e(H) : H \text{ is an induced subgraph of } G\}$$

Erdős and McKay [5] conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G).$$

It was proven for Random graphs by Calkin, Frieze, and Mckay [3], but the conjecture for Ramsey graphs remained open for decades and only started to show significant progress in recent years. In particular, a chain of results were sparked after the work of Narayanan, Sahasrabudhe, and Tomon [10] establishing that $|\Phi(G)| = n^{2-o(1)}$ for Ramsey graph G on n vertices. Their proof builds on some ideas from the work of Bukh and Sudakov [2] which showed that every Ramsey graph contains a linear sized subgraph with many distinct degrees, a feature that dense random graphs also exhibit. Kwan and Sudakov [8] later improved the bound to $|\Phi(G)| = \Omega(n^2)$ by refining the argument of Narayanan, Sahasrabudhe, and Tomon. Eventually, Kwan, Sah, Sauermann, and Sawhney [9] closed the conjecture via Fourier-analytic arguments on graph polynomials.

This dissertation will start by studying the result of Bukh and Sudakov [2] on large induced subgraphs in Ramsey graphs with many distinct degrees. This work not only tightens the connection between Ramsey graphs and random graphs, but also builds the foundation for the subsequent $|\Phi(G)| = \Omega(n^2)$ result by Kwan and Sudakov which we will present in the following chapter. Lastly, we outline the proof of the Erdős–McKay conjecture by Kwan, Sah, Sauermann, and Sawhney [9].

1.1 The anticoncentration behavior

The core results of this dissertation concerns the *anticoncentration* behavior exhibited in Ramsey graphs. In general, anticoncentration provides an upper bound on the point probabilities of a random variable. In the following chapters, we will repeatedly sample a uniformly random subset $U \subseteq V(G)$ and study the induced subgraph $G[U]$. A question to ask in this setting is how likely it is for two vertices $x, y \in V(G)$ to have the same degree in $G[U]$?

Observe that $\deg_U(x) - \deg_U(y)$ counts only vertices in the symmetric difference $D = N_G(x) \triangle N_G(y)$. Write $A = N(x) \setminus N(y), B = N(y) \setminus N(x)$. Then

$$\deg_U(x) - \deg_U(y) = |U \cap A| - |U \cap B|, \tag{1}$$

so x, y have the same degree in $G[U]$ precisely when $|U \cap A| = |U \cap B|$. Since each $v \in A$ contributes $+1$ to the difference $|U \cap A| - |U \cap B|$ with probability $1/2$ and each $v \in B$ contributes -1 with probability $1/2$, the difference is a sum of $|D| = |A| + |B|$ independent random variables, which has standard deviation $\Theta(\sqrt{n})$ when $|D| \geq \varepsilon n$

is large. The one-dimensional Erdős–Littlewood–Offord theorem then shows that the probability that $\deg_U(x) - \deg_U(y)$ equals any particular value is $O(1/\sqrt{n})$, and thus $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$.

An intuition to this $O(1/\sqrt{n})$ bound is that, in either sampling method, $\deg_U(x) - \deg_U(y)$ is approximately normally distributed with standard deviation $\Theta(\sqrt{n})$. Since the probability density function $\phi(x)$ of a normal distribution with standard deviation σ is at most $1/(\sigma\sqrt{2\pi})$ for all x, the probability that $\deg_U(x) - \deg_U(y)$ equals any particular value is $O(1/\sigma) = O(1/\sqrt{n})$, and this anticoncentration bound is the key to the results in Chapter 2 and 3. In Chapter 4, we will instead study the induced edge count $e(U)$ itself, which turns out to vaguely resemble the normal distribution and exhibit an anticoncentration behavior.

1.2 Notations and tools

Let $G = (V, E)$ be a graph. We call $e(G) := |E(G)|$ the *size* of G. For $U \subseteq V$, $G[U]$ denotes the subgraph of G induced by U, and write $e(U) := e(G[U])$.

We omit all floor and ceiling signs when they are not crucial for simplicity.

We will use the following probability bounds throughout the dissertation.

Proposition 1.5 (Markov's inequality)**.** *Let X be a non-negative random variable. Then for $t \geq 0$,*

$$\mathbb{P}\Big[X \geq t\Big] \leq \frac{\mathbb{E}[X]}{t}.$$

□

Proposition 1.6 (Chebyshev's inequality)**.** *Let X be a random variable. Then for $t > 0$,*

$$\mathbb{P}\Big[|X - \mathbb{E}[X]| \geq t\Big] \leq \frac{\mathrm{Var}(X)}{t^2}.$$

□

For $X = \mathbb{1}_{A_1} + \cdots + \mathbb{1}_{A_k}$, we often use the following inequality to bound the variance:

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \sum_i \sum_j \mathbb{P}[A_i \cap A_j] - \mathbb{P}[A_i]\mathbb{P}[A_j] \\
&= \sum_i \Big[\mathbb{P}[A_i] - \mathbb{P}[A_i]^2\Big] + \sum_i \sum_{j \sim i} \Big[\mathbb{P}[A_i \cap A_j] - \mathbb{P}[A_i]\mathbb{P}[A_j]\Big] \\
&\leq \mathbb{E}[X] + \sum_i \sum_{j \sim i} \mathbb{P}[A_i \cap A_j]
\end{aligned}$$

where we write $i \sim j$ if $i \neq j$ and A_i, A_j are dependent.

Proposition 1.7 (Chernoff bound)**.** *Let X be a binomial random variable with parameters n and p. Then for $t \geq 0$,*

$$\mathbb{P}\left[X \leq \mathbb{E}[X] - t\right], \mathbb{P}\left[X \geq \mathbb{E}[X] + t\right] \leq \exp\Big(-2t^2/n\Big).$$

□

Proposition 1.8 (Hoeffding's inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent real-valued random variables with $0 \le X_i \le 1$ for each i, and write $X = \sum_{i=1}^n X_i$. Then for $t \ge 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \ge t) \le 2e^{-2t^2/n}.$$

□

We will use the following version of Turán's theorem frequently.

Proposition 1.9 (Turán's theorem)**.** *Every graph G with n vertices and m edges contains an independent set of size at least $n^2/(2m+n)$.*

Proof. Consider a random ordering $<$ on $V(G)$. Define

$$I \coloneqq \Big\{v \in V : \{v, u\} \in E(G) \implies v < u\Big\}.$$

If $u, v \in I$ are adjacent, say $u > v$, then v does not belong in I, contradiction. Thus I is an independent set. Since $v \in I$ if and only if v is the least element among v and its neighbors,

$$\mathbb{E}\Big[|I|\Big] = \sum_{v \in V} \mathbb{P}\Big[v \in I\Big] = \sum_{v \in V} \frac{1}{1 + \deg(v)}.$$

The function $f(x) = 1/(x+1)$ is convex on $[0, \infty)$, so by Jensen's inequality, G contains an independent set I of size

$$|I| \ge \sum_{v \in V} \frac{1}{1 + \deg(v)} \ge \frac{n}{1 + \frac{1}{n}\sum_{v \in V} \deg(v)} = \frac{n^2}{2m+n}.$$

□

Definition 1.5. A random variable X is of (n,p)-*Littlewood–Offord type* if it is of the form $X = a_1\xi_1 + a_2\xi_2 + \cdots + a_n\xi_n + C$, where $a_1, a_2, \ldots, a_n \in \mathbb{Z} \setminus \{0\}$ and $C \in \mathbb{Z}$ are fixed and $\xi_1, \xi_2, \ldots, \xi_n$ are independent Bernoulli random variables with parameter p.

Proposition 1.10 (Littlewood–Offord theorem)**.** *Let $p > 0$ be constant and X be a random variable of (n,p)-Littlewood–Offord type. Then for any $t \in \mathbb{Z}$,*

$$\mathbb{P}[X = t] = O\Big(1/\sqrt{n}\Big).$$

Proof. We first consider the case where all coefficients a_i are positive. Without loss of generality, assume $X \coloneqq \sum_{i=1}^n a_i\xi_i$. Since $a_i \ge 1$ for all i, we have $1 + \sum_{i \in A} a_i \le \sum_{i \in B} a_i$ whenever $A \subset B$. For $t \in \mathbb{Z}$, let

$$\mathcal{A}_t \coloneqq \left\{A : \sum_{i \in A} a_i = t\right\} \subseteq [n],$$

so no elements of $\mathcal{A}_t$ are contained in another. Consider the chain $C = (S_1, S_2, \ldots, S_n)$ of n subsets of $[n]$ such that $\emptyset = S_1 \subset S_2 \subset \cdots \subset S_n = [n]$. There are $n!$ such chains C in $[n]$, and for $A \subseteq [n]$, there are $|A|!(n-|A|)!$ of them containing A. Since no chain can contain multiple elements of $\mathcal{A}_t$,

$$\sum_{A \in \mathcal{A}_t} |A|!(n-|A|)! \leq n! \implies \sum_{A \in \mathcal{A}_t} \frac{1}{\binom{n}{|A|}} \leq 1.$$

It follows from Stirling's formula on binomial coefficients that

$$\begin{aligned}
\mathbb{P}\Big[X = t\Big] &\leq \sum_{A \in \mathcal{A}_t} p^{|A|}(1-p)^{n-|A|} \\
&\leq \sum_{A \in \mathcal{A}_t} \frac{1}{\binom{n}{|A|}} \cdot \binom{n}{|A|} p^{|A|}(1-p)^{n-|A|} \\
&\leq \left(\sum_{A \in \mathcal{A}_t} \frac{1}{\binom{n}{|A|}} \right) \cdot \binom{n}{np} p^{np}(1-p)^{n-np} \\
&\leq \frac{1+o(1)}{\sqrt{2\pi p(1-p)n}} = O\Big(1/\sqrt{n}\Big).
\end{aligned}$$

Note that the third inequality holds by the probability mass function of the binomial distribution.

Now consider the general case. Let $I := \{i : a_i \geq 1\}$ and $J := \{i : a_i \leq -1\}$. We may assume $|I| \geq n/2$ by passing to $-X$ and swapping I with J if necessary. Then conditioning on $(\xi_j)_{j \in J}$, X is of Littlewood–Offord type with positive coefficients, so

$$\mathbb{P}\Big[X = t \mid (\xi_j)_{j \in J}\Big] = O\Big(1/\sqrt{|I|}\Big) = O\Big(1/\sqrt{n}\Big).$$

The tower property of conditional expectation gives

$$\mathbb{P}\Big[X = t\Big] = \mathbb{E}\left[\mathbb{P}\Big[X = t \mid (\xi_j)_{j \in J}\Big]\right] = O\Big(1/\sqrt{n}\Big).$$

$\square$

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [2])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [5] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely. In particular, this theorem already has implications for the edge statistics of Ramsey graphs G, as the subgraphs $G - \{v\}$ for $v \in V(G)$ have $\Omega(\sqrt{n})$ different sizes. This idea lays the foundation for the proof in the next chapter.

To prove Theorem 2.1, Bukh and Sudakov introduced the notion of *diversity*, which captures the idea that the neighbourhoods of most pairs of vertices are not too similar.

Definition 2.1. An n-vertex graph G is (ε, δ)-*diverse* if for $x \in V(G)$, we have $|N(x) \triangle N(y)| \geq \varepsilon n$ for $\leq n^{\delta}$ vertices $y \in V(G)$.

Notice that (ε, δ)-diversity directly conditions our assumption that $|N(x) \triangle N(y)| = D \geq \varepsilon n$ in the anticoncentration principle discussion in Section 1.1. In particular, we may show in a diverse graph that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$ for all but $\leq n^{\delta}$ pairs of vertices $x, y \in V(G)$, from which we use Turan's theorem to conclude that there are many distinct degrees in a diverse graph. The following lemma formalizes this observation and reduces Theorem 2.1 to showing every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.2. *Let $\delta \in (0, 1/2)$ and let G be an (ε, δ)-diverse graph on n vertices for large enough n. Then there is a subgraph on $\geq n/4$ vertices containing vertices of $\Omega(\sqrt{n})$ distinct degrees.*

Proof. Let U be a uniformly random subset of $V(G)$. It suffices to show that with positive probability, $|U| \geq n/4$ and $G[U]$ contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in U$ and $\deg_U(x) = \deg_U(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \leq \mathbb{P}[\deg_U(x) = \deg_U(y)]$. Put $D = N_G(x) \triangle N_G(y), A = N_G(x) \setminus N_G(y)$, and $B = N_G(y) \setminus N_G(x)$. Then the random variable

$$\deg_U(x) - \deg_U(y) = |A \cap U| - |B \cap U| = \sum_{v \in A} \mathbb{1}_{v \in U} - \sum_{v \in B} \mathbb{1}_{v \in U}$$

is of $(|D|, 1/2)$-Littlewood–Offord type. Thus by Littlewood–Offord theorem (Proposition 1.10),

$$\mathbb{P}\Big[\deg_U(x) = \deg_U(y)\Big] = O\Big(1/\sqrt{|D|}\Big).$$

In particular, $\mathbb{E}[\mathbb{1}_{xy}] = O(1/\sqrt{n})$, whenever $|D| \geq \varepsilon n$.

Now consider the degree collision graph H on U, where $x, y \in U$ are adjacent if and only if $\deg_U(x) = \deg_U(y)$. Since G is (ε, δ)-diverse, $\mathbb{E}[\mathbb{1}_{xy}] = O(1/\sqrt{n})$ for all but $\leq n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Hence, the expected number of edges in H is

$$\mathbb{E}[e(H)] = \sum_{\{x,y\} \in V(G)^2} \mathbb{E}[\mathbb{1}_{xy}] \leq n^{1+\delta} + n^2 \cdot O\Big(1/\sqrt{n}\Big) = O\Big(n^{3/2}\Big)$$

and Markov's inequality yields $\mathbb{P}[e(H) = O(n^{3/2})] \geq 1/2$. Since $|U| < n/4$ with probability $o(1)$ by the Chernoff bound, the union bound gives a positive probability that $|U| \geq n/4$ and $e(H) \leq O(n^{3/2})$ occur simultaneously, in which case Turan's Theorem (Proposition 1.9) gives an independent set I in H of size at least

$$|I| \geq \frac{|U|^2}{2e(H) + |U|} = \frac{(n/4)^2}{O(n^{3/2}) + n/4} = \Omega\Big(\sqrt{n}\Big).$$

In particular, each vertex in I has a distinct degree in $G[U]$. □

It remains to show that every Ramsey graph contains a diverse subgraph of linear size. Instead of directly proving this, we prove the same statement on a strengthening of diversity called *richness*.

Definition 2.2. An n-vertex graph G is (ε, δ)-*rich* if, for $W \subseteq V(G)$ with $|W| \geq \delta n$, we have $|N(v) \cap W| \geq \varepsilon |W|$ and $|\overline{N(v)} \cap W| \geq \varepsilon |W|$ for $\leq n^\delta$ vertices $v \in V(G)$.

The key property of richness is that it ensures the intersection of vertex neighbourhoods with arbitrary large vertex sets are not extremely large or small. When W is the neighborhood of another vertex, richness ensures both sides of the symmetric difference $|N(x) \triangle N(y)|$ are large, rather than merely the symmetric difference itself being large, which is why richness is a stronger condition than diversity.

Lemma 2.3. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then G is $(\varepsilon/2, \delta)$-diverse.*

Proof. Note that for each $x \in V(G)$, either $|N(x)| \geq n/2$ or $|\overline{N(x)}| \geq n/2$. If $|N(x)| \geq n/2$, then richness implies that

$$|\overline{N(y)} \cap N(x)| \geq \varepsilon |N(x)| \geq \varepsilon n/2$$

for $\leq n^\delta$ vertices $y \in V(G)$. Similarly, if $|\overline{N(x)}| \geq n/2$, then

$$|N(y) \cap \overline{N(x)}| \geq \varepsilon |\overline{N(x)}| \geq \varepsilon n/2$$

for $\leq n^\delta$ vertices $y \in V(G)$. But then in either case,

$$|N(x) \triangle N(y)| = |\overline{N(y)} \cap N(x)| + |N(y) \cap \overline{N(x)}| \geq \varepsilon n/2.$$

□

We now show that every Ramsey graph contains a rich subgraph of linear size.

Lemma 2.4. *For $\delta \in (0,1)$, there exists $\varepsilon = \varepsilon(C,\delta) > 0$ such that every C-Ramsey graph G on n vertices contains an (ε,δ)-rich subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = K(C)$ be a large constant to be determined. Let $c = 4^{-K}$ and $\varepsilon = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size $s = (cn)^\delta/2$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 4\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 4\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G[U_0] \supset G[U_1] \supset \cdots \supset G[U_K]$, such that $|U_i| \geq (\delta/4)|U_{i-1}|$ and $S_i \subseteq U_{i-1}$ for all i.

We first show how this construction gives the lemma. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size s and $\varepsilon = 1/K$,

$$d(S) = \frac{\sum_{k=1}^r e(S_{i_k})}{\binom{|S|}{2}} + \frac{\sum_{k<l} e(S_{i_k}, S_{i_l})}{\binom{|S|}{2}} \leq \frac{r\binom{s}{2}}{\binom{rs}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 4\varepsilon \leq \frac{6}{K}.$$

The Erdős–Szemerédi theorem (Theorem 1.3) now yields a homogeneous set $A \subseteq S$ of size

$$|A| \geq \frac{\gamma K}{10\log(K/10)} \log|S| = \frac{\gamma\delta K}{10\log(K/10)} \log cn = \frac{\gamma\delta K}{10\log(K/10)} \log n + o(\log n),$$

for some absolute constant $\gamma > 0$ given by the theorem. Take K to be large enough so that $\frac{\gamma\delta K}{10\log(K/10)} > C$. Then $|A| > C\log n$, contradiction.

We now construct the sets S_i and U_i inductively. Suppose $U_1, \ldots, U_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|U_{i-1}| \geq (\delta/4)^{i-1}|U_0| \geq cn$. Since $G[U_{i-1}]$ is not (ε,δ)-rich by assumption, U_{i-1} contains subsets W_i, Y_i with $|W_i| \geq \delta|U_{i-1}|$ and $|Y_i| = (cn)^\delta$, such that for each $y \in Y_i$ we have $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ or $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon|W_i|$. We consider two cases.

Suppose $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ for half the vertices $y \in Y_i$, and let S_i be the corresponding subset of Y_i, so $|S_i| = (cn)^\delta/2$. Put $T_i = W_i \setminus S_i$ and note that $|T_i| \geq |W_i|/2$, as $|S_i| \leq |W_i|/2$ for large enough n. Then for $y \in S_i$,

$$d(y, T_i) = \frac{|N_{G_{i-1}}(y) \cap T_i|}{|T_i|} \leq \frac{|N_{G_{i-1}}(y) \cap W_i|}{|T_i|} < \frac{\varepsilon|W_i|}{|W_i|/2} = 2\varepsilon.$$

Partition T_i into $U_i \sqcup Z_i$ where $U_i = \{v \in T_i : d(v, S_i) \leq 4\varepsilon\}$. We show that $|U_i| \geq (\delta/4)|U_{i-1}|$ by double counting $e(T_i, S_i)/|S_i|$. Since $d(v, S_i) > 4\varepsilon$ for $v \in Z_i$,

$$\frac{e(T_i, S_i)}{|S_i|} = \sum_{v \in T_i} d(v, S_i) \geq \sum_{v \in Z_i} d(v, S_i) > 4\varepsilon|Z_i|.$$

On the other hand, since $d(y, T_i) < 2\varepsilon$ for $y \in S_i$,

$$\frac{e(T_i, S_i)}{|S_i|} = \frac{1}{|S_i|} \sum_{y \in S_i} e(y, T_i) = \frac{|T_i|}{|S_i|} \sum_{y \in S_i} d(y, T_i) < 2\varepsilon |T_i|.$$

Combining both inequalities yields $|Z_i| < |T_i|/2$. Therefore

$$|U_i| = |T_i| - |Z_i| > \frac{|T_i|}{2} \geq \frac{|W_i|}{4} \geq \frac{\delta}{4} |U_{i-1}|.$$

For $j > i$, since $S_j \subseteq U_j \subseteq U_i$, we have $d(v, S_i) \leq 4\varepsilon$ for every $v \in S_j$, and averaging gives $d(S_i, S_j) \leq 4\varepsilon$. Thus S_i satisfies condition (i).

Otherwise, $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon |W_i|$ for half the vertices $y \in Y_i$. Let S_i be the corresponding subset of Y_i and put $T_i = W_i \setminus S_i$, so again $|T_i| \geq |W_i|/2$ and $|S_i| = (cn)^\delta/2$. For $y \in S_i$,

$$1 - d(y, T_i) = \frac{|\overline{N_{G_{i-1}}(y)} \cap T_i|}{|T_i|} \leq \frac{|\overline{N_{G_{i-1}}(y)} \cap W_i|}{|T_i|} < \frac{\varepsilon |W_i|}{|W_i|/2} = 2\varepsilon,$$

so $d(y, T_i) > 1 - 2\varepsilon$ for every $y \in S_i$. Partition T_i into $U_i \sqcup Z_i$ where $U_i = \{v \in T_i : d(v, S_i) \geq 1 - 4\varepsilon\}$. We double count $|T_i| - e(T_i, S_i)/|S_i| = \sum_{v \in T_i} (1 - d(v, S_i))$. Since $1 - d(v, S_i) > 4\varepsilon$ for $v \in Z_i$,

$$|T_i| - \frac{e(T_i, S_i)}{|S_i|} = \sum_{v \in T_i} (1 - d(v, S_i)) \geq \sum_{v \in Z_i} (1 - d(v, S_i)) > 4\varepsilon |Z_i|.$$

On the other hand, using $d(y, T_i) > 1 - 2\varepsilon$ for $y \in S_i$,

$$|T_i| - \frac{e(T_i, S_i)}{|S_i|} = |T_i| - \frac{|T_i|}{|S_i|} \sum_{y \in S_i} d(y, T_i) < |T_i| - (1 - 2\varepsilon)|T_i| = 2\varepsilon |T_i|.$$

Combining both inequalities gives $|Z_i| < |T_i|/2$, hence $|U_i| > |T_i|/2 \geq (\delta/4)|U_{i-1}|$ as in the other case. For $j > i$, each $v \in S_j \subseteq U_i$ satisfies $d(v, S_i) \geq 1 - 4\varepsilon$, so $d(S_i, S_j) \geq 1 - 4\varepsilon$. Thus S_i satisfies condition (ii). This completes the induction. □

Theorem 2.1 now follows by a simple application of Lemmas 2.4, 2.3, and 2.2 in sequence.

Proof of Theorem 2.1. Let G be a C-Ramsey graph on n vertices, and fix any $\delta \in (0, 1/2)$. By Lemma 2.4, there exists an (ε, δ)-rich induced subgraph $G' \subseteq G$ on $m = \Omega(n)$ vertices. By Lemma 2.3, G' is $(\varepsilon/2, \delta)$-diverse. Applying Lemma 2.2 to G', we obtain an induced subgraph of G' on $\Omega(m) = \Omega(n)$ vertices containing $\Omega(\sqrt{m}) = \Omega(\sqrt{n})$ vertices of distinct degrees. □

3 Subgraphs of Quadratically Many Sizes

Erdős and Mckay conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G) := \{e(H) : H \text{ is an induced subgraph of } G\},$$

That is, for any $1 \leq t \leq \delta n^2$, there exists an induced subgraph with exactly t edges. The conjecture was widely open for decades. In 2003, Alon, Krivelevich and Sudakov [1] proved the conjecture with n^δ in place of δn^2, but little progress was made afterwards.

Theorem 3.1 (Alon, Krivelevich and Sudakov [1])**.** *For any $C > 0$, there exists $\delta = \delta(C) > 0$ such that*

$$\{1, 2, \ldots, n^\delta\} \subseteq \Phi(G).$$

Since the conjecture implies $|\Phi(G)| = \Omega(n^2)$ and the triavial upper bound gives $|\Phi(G)| \leq \binom{n}{2}$, a natural relaxation is to ask whether $\Phi(G)$ indeed has quadratic size. In 2017, Narayanan, Sahasrabudhe, and Tomon [10] proved a near optimal bound of $|\Phi(G)| = n^{2-o(1)}$ by building on the notion of diversity from Chapter 2. The desired quadratic bound was achieved by Kwan and Sudakov [8] a few years later, and this chapter is dedicated to presenting their proof.

Theorem 3.2 (Kwan and Sudakov [8])**.** *For any $C > 0$,*

$$|\Phi(G)| = \Theta(n^2).$$

Recall that Theorem 2.1 provides a bound of $|\Phi(G)| = \Omega(\sqrt{n})$ for any Ramsey graph G by removing a single vertex to construct each subgraph. Two ideas may be used to improve this construction. First, we apply the same idea to many different induced subgraphs rather than merely G itself. Second, instead of deleting only one vertex, we add and remove various combinations of vertices to produce many more distinct sizes.

Specifically, pick a target m and find two disjoint vertex sets U and W, where U is a linear-sized random set with $e(U) \approx m$ and W is an augmenting set of size $O(\sqrt{n})$ selected from a candidate set $W_0 \subseteq V(G) \setminus U$ whose degrees are of linear order and all lie in a narrow interval. Then the subsets $Z \subseteq W$ produce $\Omega(n^{3/2})$ distinct values of $e(U \cup Z)$ that all lie in an interval of length $O(n^{3/2})$ centred at $e(U)$. Applying this to $\Omega(\sqrt{n})$ well-separated targets m yields the full $\Omega(n^2)$ count.

Kwan and Sudakov construct W as three disjoint parts $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$, each of size $\Theta(\sqrt{n})$, that act at complementary scales. Every element has $\Theta(n)$ edges into U, and the degrees from $\mathbf{S}$ into U are uniformly smaller than those from $\mathbf{T}$ by a gap of $\Omega(\sqrt{n})$. Varying the number k of elements of $\mathbf{S}$ included in Z shifts $e(U \cup Z)$ at the coarsest scale $\Theta(n)$. Exchanging elements of $\mathbf{S}$ for elements of $\mathbf{T}$ shifts it by $\Theta(\sqrt{n})$. The elements of $\mathbf{X}$ fill the remaining $\Theta(\sqrt{n})$ gap by contributing $\Theta(\sqrt{n})$ distinct degrees

into $U \cup Z$ for each choice of Z. Hence the different combinations of $\mathbf{S}, \mathbf{T}, \mathbf{X}$ yield $\Omega(n^{3/2})$ distinct values of $e(U \cup Z)$.

Achieving the requirement for $\mathbf{X}$ is the main technical challenge, as vertex diversity defined in Chapter 2 does not produce enough distinct degrees in this case. By the anticoncentration principle from Section 1.1, $\mathbb{P}[d_U(x) = d_U(y)] = O(1/\sqrt{n})$ for all but $\leq n^\delta$ pairs $x, y \in V(G)$ in an (ε, δ)-diverse graph. Following the same construction as in the proof of Lemma 2.2, the collision graph H on the candidate set W_0 has expected number of edges

$$\mathbb{E}[e(H)] = O(|W_0|^2/\sqrt{n} + |W_0| \cdot n^\delta),$$

which yields an independent set $I \subseteq W_0$ of size

$$|I| = \frac{|W_0|^2}{2\mathbb{E}[e(H)] + |W_0|} = \Omega\left(\sqrt{n} \cdot \frac{|W_0|}{|W_0| + n^{1/2+\delta}}\right).$$

by Turán's theorem (Proposition 1.9). We thus need $|W_0| = \Omega(n^{1/2+\delta})$ for $|I|$ to reach $\Omega(\sqrt{n})$. Unfortunately, W_0 is extracted by pigeonholing n vertices into $O(\sqrt{n})$ degree intervals of length $O(\sqrt{n})$, so at best $|W_0| = \Theta(\sqrt{n})$, yielding only $|I| = \Omega(n^{1/2-\delta})$.

To overcome this obstacle, Kwan and Sudakov's allows the canditate set W_0 to also contain disjoint *pairs*. Write $\mathbf{x}$ in bold denote an object that is either a single vertex or a pair of vertices, and similarly for sets of such objects. For $\mathbf{x} = \{x_1, x_2\}$, set $N(\mathbf{x}) := N(x_1) \cup N(x_2)$ as a multiset union and $d(\mathbf{x}) := |N(\mathbf{x})| = d(x_1) + d(x_2)$ as the pair-degree. Note that there are $\binom{n}{2} = \Theta(n^2)$ pairs but the pair-degrees take only $O(n)$ values. Pigeonholing now yields $\Theta(n)$ disjoint pairs sharing a common pair-degree, vastly exceeding the $n^{1/2+\delta}$ threshold. The price is that we need the symmetric difference $|N(\mathbf{x}) \triangle N(\mathbf{y})| = \Omega(n)$ not just for most pairs of vertices, but for most pairs of *pairs*. We call this *pair-diversity*.

Definition 3.1. An n-vertex graph G is $(\varepsilon, \delta, \alpha)_2$-*diverse* if for each pair of vertices $\mathbf{x} = \{x_1, x_2\}$ in G with $|N(x_1) \triangle N(x_2)| \geq \alpha n$, there are $\leq n^\delta$ disjoint pairs of vertices $\mathbf{y} = \{y_1, y_2\}$ with $\mathbf{x} \cap \mathbf{y} = \emptyset$, such that $|N(\mathbf{x}) \triangle N(\mathbf{y})| < \varepsilon n$.

It turns out that richness also implies pair-diversity.

Lemma 3.3. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then G is $(\varepsilon, \delta, \alpha)_2$-diverse for any $\alpha \geq 2\delta$.*

Proof. Suppose not. Then there exists a pair $\mathbf{x} = \{x_1, x_2\}$ with $|N(x_1) \triangle \overline{N(x_2)}| \geq \alpha n$ and a set Y of n^δ pairs such that $|N(\mathbf{x}) \triangle N(\mathbf{y})| < \alpha\varepsilon n/2$ for all $\mathbf{y} \in Y$. Since

$$|N(x_1) \triangle \overline{N(x_2)}| = |N(x_1) \cap N(x_2)| + |\overline{N(x_1)} \cap \overline{N(x_2)}| \geq \alpha n,$$

we have $|N(x_1) \cap N(x_2)| \geq (\alpha/2)n \geq \delta n$ or $|\overline{N(x_1)} \cap \overline{N(x_2)}| \geq (\alpha/2)n \geq \delta n$. Assume without loss of generality that the former case holds. Then for each vertex y in any $\mathbf{y} \in Y$,

$$\left|\overline{N(y)} \cap (N(x_1) \cap N(x_2))\right| \leq |N(\mathbf{x}) \triangle N(\mathbf{y})| < \alpha\varepsilon n/2 \leq \varepsilon |N(x_1) \cap N(x_2)|.$$

But then the set of all such y now contradicts the (ε, δ)-rich condition. $\square$

We also introduce a reversed notion of diversity.

Lemma 3.4. *Let an n-vertex graph G be (ε, δ)-rich. Then G has $\leq n^{1+\delta}$ pairs of vertices $x_1, x_2 \in V(G)$ with $|N(x_1) \triangle \overline{N(x_2)}| < (\varepsilon/2)n$.*

Proof. We show that for each choice of $x_1 \in V(G)$, there are $\leq n^{\delta}$ vertices $x_2 \in V(G)$ with $|N(x_1) \triangle \overline{N(x_2)}| < (\varepsilon/2)n$. Since either $|N(x_1)| \geq n/2$ or $|\overline{N(x_1)}| \geq n/2$, we assume without loss of generality that $|N(x_1)| \geq n/2$. By (ε, δ)-richness, there are $\leq n^{\delta}$ vertices x_2 with $|N(x_2) \cap N(x_1)| < \varepsilon n/2$. For all other choices of x_2,

$$|N(x_1) \triangle \overline{N(x_2)}| \geq |N(x_1) \cap N(x_2)| \geq (\varepsilon/2)n.$$

The result now follows from the fact that there are only n choices of x_1. $\square$

Remark: This lemma gives an upper bound on the symmetric difference between the neighbourhoods of two vertices. Specifically, since $|N(x_1) \triangle N(x_2)| + |N(x_1) \triangle \overline{N(x_2)}| = n - 2$ for any two vertices x_1, x_2,

$$|N(x_1) \triangle N(x_2)| \leq (1 - \varepsilon/2)n$$

for most pairs of vertices x_1, x_2, so combined with diversity yields $d(x_1) + d(x_2) = \Theta(n)$.

One last thing we need is to ensure $\mathbf{X}$ actually provides $\Theta(\sqrt{n})$ distinct degrees into $U \cup \mathbf{Z} \coloneqq U \cup \bigcup_{\mathbf{z} \in \mathbf{Z}} \mathbf{z}$ for each choice of augmenting set $\mathbf{Z} \subseteq \mathbf{S} \cup \mathbf{T}$. Notice the current construction of $\mathbf{X}$ only ensures distinct degrees $d_U(\mathbf{x})$ into U instead of $d_{U \cup Z}(\mathbf{x})$. If U were sampled once and directly constructed $\mathbf{S}, \mathbf{T}, \mathbf{X}$ based on U, then the $d_U(\mathbf{x})$ would be fixed numbers given the outcome of U. However, there is no guarantee that $d_{U \cup Z}(\mathbf{x}) = d_U(\mathbf{x}) + d_Z(\mathbf{x})$ would remain distinct, as the addition of $d_Z(\mathbf{x})$ could cancel out the uniqueness of $d_U(\mathbf{x})$. Kwan and Sudakov overcome this with a *double-exposure* technique. They perform two rounds of sampling. First, they sample the random set U_0 and construct $\mathbf{S}, \mathbf{T}, \mathbf{X}$ relative to U_0. In the second round, they sample U from U_0 by independently keeping each vertex with probability $1/2$. After the first sampling, the construction guarantees $|N_{U_0}(\mathbf{x}) \triangle N_{U_0}(\mathbf{y})| = \Omega(n)$ for all $x, y \in \mathbf{X}$, and the augmenting sets $\mathbf{Z} \subseteq \mathbf{S} \cup \mathbf{T}$ are fully determined by U_0. In the second round, the difference $d_U(\mathbf{x}) - d_U(\mathbf{y})$ becomes a sum of $\Omega(n)$ independent Bernoulli terms, and Littlewood–Offord yields $\mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t] = O(1/\sqrt{n})$ for every integer t. In particular, since $t_{xy} \coloneqq d_Z(\mathbf{y}) - d_Z(\mathbf{x})$ is determined for each $\mathbf{x}, \mathbf{y}$,

$$\mathbb{P}[d_{U \cup Z}(\mathbf{x}) = d_{U \cup Z}(\mathbf{y})] = \mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t_{xy}] = O(1/\sqrt{n}).$$

The same collision graph construction now finds us the desired set of U such that $d_{U \cup Z}(\mathbf{x})$ is distinct for $\mathbf{x} \in \mathbf{X}$.

3.1 The main lemma

As outlined above, we will find many induced subgraphs $G[U]$ whose edge counts are well-separated from each other, and show that each can be augmented in $\Omega(n^{3/2})$ ways. Theorem 3.2 will be an immediate consequence of the following lemma, which we will dedicate the rest of this chapter to prove.

Lemma 3.5. *Let $C > 0$. Then there exists $c = c(C) > 0$, such that for any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \leq m \leq 2cn^2$, there are disjoint subsets $U, W \subseteq V(G)$ with $|e(U) - m| = O(n^{3/2})$ and $|W| = O(\sqrt{n})$ such that*

$$|\{e(U \cup Z) : Z \subseteq W\}| = \Omega(n^{3/2}).$$

We first show how this implies Theorem 3.2.

Proof of Theorem 3.2. For any U, W given by Lemma 3.5,

$$|e(U \cup W) - e(U)| = O(n^{3/2}),$$

as $|W| = O(\sqrt{n})$ and each $w \in W$ has $\leq n$ neighbours in $U \cup W$. But then the $\Omega(n^{3/2})$ values of $e(U \cup Z)$ for $Z \subseteq W$ are contained in an interval of length $O(n^{3/2})$ centered at $e(U)$. Partition the interval between cn^2 and $2cn^2$ into $\frac{c}{k}\sqrt{n}$ subintervals of length $kn^{3/2}$, for some large enough constant k to be determined. Apply Lemma 3.5 to a suitable choice of m in each subinterval so that the obtained values of $e(U \cup Z)$ for $Z \subseteq W$ lie in the same subinterval as m, which is allowed for sufficiently large k. Since we applied Lemma 3.5 to $\Omega(\sqrt{n})$ disjoint subintervals, we obtain $\Omega(n^2)$ distinct sizes of induced subgraphs. □

3.2 The construction of augmenting sets

The following lemma constructs U_0 and the augmenting sets $\mathbf{S}, \mathbf{T}, \mathbf{X}$ with the properties needed for Lemma 3.5.

Lemma 3.6. *For any $C > 0$ there exists $c = c(C) > 0$ such that the following holds. For any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \leq m \leq 2cn^2$, there is a vertex set $U_0 \subseteq V(G)$ with $|e(U_0) - 4m| = O(n^{3/2})$, a constant $c' = c'(C) > 0$, and sets $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$ of size $c'\sqrt{n}$ such that*

1. *either $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq V(G)$ are disjoint sets of vertices, or $\mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq \binom{V(G)}{2}$ are sets of disjoint pairs of vertices with no vertex appearing in more than one of $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}$,*
2. *there is $d = \Theta(n)$ such that $d_{U_0}(\mathbf{x}) = d + O(\sqrt{n})$ for each $\mathbf{x} \in \mathbf{S} \cup \mathbf{T} \cup \mathbf{X}$,*
3. *$\min_{\mathbf{x} \in \mathbf{T}} d_{U_0}(\mathbf{x}) - \max_{\mathbf{x} \in \mathbf{S}} d_{U_0}(\mathbf{x}) \geq 8c'\sqrt{n}$,*
4. *for each $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{X}}{2}$, we have $|N_{U_0}(\mathbf{x}) \triangle N_{U_0}(\mathbf{y})| = \Omega(n)$.*

Property 2 ensures adding or removing a single element shifts the edge count by $\Theta(n)$. Property 3 guarantees swapping an element of $\mathbf{S}$ for an element of $\mathbf{T}$ shifts the edge count by $\Theta(\sqrt{n})$. Property 4 sets up the double-exposure step.

Proof of Lemma 3.6. Let $\varepsilon = \varepsilon(C)$ be given by Lemma 2.4 and set $\delta = \varepsilon/4$. By Lemma 2.4, G contains an (ε, δ)-rich induced subgraph $G[V']$ on $\Omega(n)$ vertices. Since $\log |V'| \geq \frac{1}{2} \log n$, the graph $G[V']$ is $2C$-Ramsey, so by adjusting constants it suffices to work inside $G[V']$. We may thus assume G itself is (ε, δ)-rich.

By richness, Lemma 2.3, 3.3, and 3.4 with $\alpha = \varepsilon/2$, furnishes the following properties:

(a) G is $(\varepsilon/2, \delta)$-diverse,

(b) G is $(\varepsilon^2/4, \delta, \varepsilon/2)_2$-diverse,

(c) there are $\leq n^{1+\delta}$ pairs $\{x_1, x_2\}$ with $|N(x_1) \triangle \overline{N(x_2)}| < (\varepsilon/2)n$.

Each of the $\binom{n}{2} = \Theta(n^2)$ vertex pairs $\{x_1, x_2\}$ has a pair-degree $d(\{x_1, x_2\}) = d(x_1) + d(x_2)$ taking values from 0 to $2n$, so by the pigeonhole principle there exists a value d^* and a collection H of $\Omega(n^{3/2})$ pairs with $d(\mathbf{x}) = d^* + O(\sqrt{n})$ for all $\mathbf{x} \in H$. By the remark after Lemma 3.4, $d^* = \Theta(n)$. Interpret H as a graph on $V(G)$ whose edges are the $\Omega(n^{3/2})$ pairs. By (c), we may delete $O(n^{1+\delta}) = o(n^{3/2})$ edges in H so that all remaining pairs have $|N(x_1) \triangle \overline{N(x_2)}| \geq (\varepsilon/2)n$. Suppose there exists vertex v in H with degree $\Omega(n^{3/4})$. Take $\mathbf{W}_0 = N_H(v)$ and set $d = d^* - d(v)$. Then each $w \in \mathbf{W}_0$ is a single vertex with degree $d(w) = d + O(\sqrt{n})$. Now suppose H has maximum degree $\Delta = o(n^{3/4})$. By iteratively picking an edge and removing $\leq 2\Delta$ edges adjacent to it, we obtain a matching of size $\Omega(n^{3/2})/o(n^{3/4}) = \Omega(n^{3/4})$. Take $\mathbf{W}_0$ to be the pairs in this matching and set $d = d^*$. In either case, the subsets of $\mathbf{W}_0$ satisfy Property 1-2.

Next, we extract a subset $\mathbf{A} \subseteq \mathbf{W}_0$ of size $\Omega(\sqrt{n})$ with pairwise large symmetric differences. Define graph F on $\mathbf{L}$ by joining $\mathbf{x}$ and $\mathbf{y}$ whenever $|N(\mathbf{x}) \triangle N(\mathbf{y})| < (\varepsilon^2/4)n$. By diversity, the graph F has $|\mathbf{W}_0| = \Omega(n^{3/4})$ vertices and maximum degree at most $n^\delta < n^{1/4}$. Thus by Turán's theorem, F has independent set $\mathbf{A} \subseteq \mathbf{W}_0$ with size $\Omega(|\mathbf{W}_0|/n^{1/4}) = \Omega(\sqrt{n})$. That is, for $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$, we have $|N(\mathbf{x}) \triangle N(\mathbf{y})| = \Omega(n)$.

By the Erdős–Szemerédi bound (Corollary 1.4), $e(G) \geq 800cn^2$ for some constant $c = c(C) > 0$. For $cn^2 \leq m \leq 2cn^2$, set $p = \sqrt{4m/e(G)}$, so that $p = \Omega(1)$ and $p \leq 0.1$. Let U_0 be a random subset of $V(G)$ by including each vertex with probability p.

Claim 3.7. *The following five events each hold with probability greater than* $4/5$*, and therefore all hold simultaneously with positive probability by the union bound.*

(i) $|e(U_0) - 4m| = O(n^{3/2})$,

(ii) There is a set $\mathbf{Q} \subseteq \mathbf{A}$ *involving no vertices of* U_0*, with* $|\mathbf{Q}| \geq (2/3)|\mathbf{A}|$,

(iii) There is a set $\mathbf{R} \subseteq \mathbf{A}$ *with* $|\mathbf{R}| \geq (2/3)|\mathbf{A}|$ *and* $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ *for each* $\mathbf{x} \in \mathbf{R}$,

(iv) $|N_{U_0}(\mathbf{x}) \triangle N_{U_0}(\mathbf{y})| = \Omega(n)$ *for each* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$,

(v) The equality $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ *holds for* $O(\sqrt{n})$ *pairs* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$.

Proof of Claim. We prove each part in order.

(i) Note that $\mathbb{E}[e(U_0)] = 4m$ and there are $O(n^3)$ pairs of edges in G whose presence are dependent in $G[U_0]$, as this occurs only if two edges share a vertex. For $e \in E(G)$, let A_e be the event that e is present in $G[U_0]$. Since

$$\mathrm{Var}(e(U_0)) \leq \mathbb{E}[e(U_0)] + \sum_{e \sim e'} \mathbb{P}(A_e \cap A_{e'}) = O(n^3),$$

Chebyshev's inequality yields

$$\mathbb{P}[|e(U_0) - 4m| = \omega(n^{3/2})] \leq \frac{O(n^3)}{\omega(n^3)} \to 0,$$

as $n \to \infty$.

(ii) Let $\mathbf{Q}$ be the subset of $\mathbf{A}$ involving no vertices of U_0. Then $|\mathbf{Q}|$ follows a binomial distribution with $|\mathbf{A}|$ trials, which has mean at least $(1-p)^2|\mathbf{A}| \geq 0.8|\mathbf{A}|$ and variance $O(|\mathbf{A}|)$. By Chebyshev's,

$$\mathbb{P}\left[|\mathbf{Q}| < \frac{2}{3}|\mathbf{A}|\right] \leq \mathbb{P}\left[|\mathbf{Q}| - \mathbb{E}[|\mathbf{Q}|] \geq 0.1|\mathbf{A}|\right] \leq \frac{O(|\mathbf{A}|)}{0.01|\mathbf{A}|^2} = O(1/|\mathbf{A}|) \to 0,$$

as $n \to \infty$.

(iii) For $\mathbf{x} \in |\mathbf{A}|$, the degree $d_{U_0}(\mathbf{x})$ has mean $pd + O(\sqrt{n})$ and variance $O(n)$. By Chebyshev's,

$$\mathbb{P}[d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})] \leq 1 - \mathbb{P}[|d_{U_0}(\mathbf{x}) - \mathbb{E}[d_{U_0}(\mathbf{x})]| = \omega(\sqrt{n})] \leq 1 - \frac{O(n)}{\omega(n)} \to 1.$$

In particular, $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ with probability 0.99 for large enough n. Let $\mathbf{R} \subseteq \mathbf{A}$ be the set of $\mathbf{x}$ that satisfies this bound. Since $\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|] = 0.01|\mathbf{A}|$, Markov's inequality gives

$$\mathbb{P}\left[|\mathbf{R}| \geq \frac{2}{3}|\mathbf{A}|\right] = 1 - \mathbb{P}\left[|\mathbf{A} \setminus \mathbf{R}| \geq \frac{1}{3}|\mathbf{A}|\right] \geq 1 - \frac{\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|]}{\frac{1}{3}|\mathbf{A}|} \geq 0.8.$$

(iv) Recall that $|N(\mathbf{x}) \triangle N(\mathbf{y})| = \Omega(n)$ for $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$. Since $|N_{U_0}(\mathbf{x}) \triangle N_{U_0}(\mathbf{y})| = |N(\mathbf{x}) \triangle N(\mathbf{y}) \cap U_0|$, it has a binomial distribution with parameters $N_{U_0}(\mathbf{x})$ and p. By the Chernoff bound,

$$\mathbb{P}\left[|N_{U_0}(\mathbf{x}) \triangle N_{U_0}(\mathbf{y})| < \frac{p}{2}|N(\mathbf{x}) \triangle N(\mathbf{y})|\right] = e^{-\Omega(n)} = o(n).$$

Union bound now gives the property.

(v) For $\{\mathbf{x},\mathbf{y}\} \in \binom{\mathbf{A}}{2}$, the random variable $d_{U_0}(\mathbf{x}) - d_{U_0}(\mathbf{y})$ is of $(|N(\mathbf{x}) \triangle N(\mathbf{y})|, p)$-Littlewood–Offord type. Since $|N(\mathbf{x}) \triangle N(\mathbf{y})| = \Omega(n)$, the Littlewood–Offord theorem gives $\mathbb{P}[d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})] = O(1/\sqrt{n})$, so the expected number of paris $\{\mathbf{x},\mathbf{y}\} \in \binom{\mathbf{A}}{2}$ satisfying $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ is $O(\sqrt{n})$. The property now follows from Markov's.

□

Fix an outcome of U_0 for which all five events hold. Let $\mathbf{A}' = \mathbf{Q} \cap \mathbf{R}$, and note that $|\mathbf{A}'| \geq |\mathbf{A}|/3 = \Omega(\sqrt{n})$. It remains to obtain $\mathbf{S}$, $\mathbf{T}$, and $\mathbf{X}$ from $\mathbf{A}'$ that satisfy Properties 3–4. The collision graph construction on $\mathbf{A}'$ and Turán Theorem (Proposition 1.9) yields $\mathbf{W} \subseteq \mathbf{A}'$ of size $\Omega(\sqrt{n})$ that has distinct degrees into U_0. Partition $\mathbf{W}$ into three equal parts by the degree $d_{U_0}(\mathbf{x})$ into U_0, and let $\mathbf{T}$ be the top third, $\mathbf{S}$ be the bottom third, and $\mathbf{X}$ be the middle third. Since $\mathbf{W}$ has distinct degrees and $|\mathbf{X}| = \Omega(\sqrt{n})$, we may find constant c' that satisfies Property 3. Since $\mathbf{X} \subseteq \mathbf{A}$, Property 4 holds. Trimming the sizes of $\mathbf{S}, \mathbf{T}, \mathbf{X}$ to $c'\sqrt{n}$ completes the proof. □

3.3 Double-exposure

Given $\mathbf{S}, \mathbf{T}, c'$ from Lemma 3.6, fix an ordering on $\mathbf{S}$ and $\mathbf{T}$ and define

$$\mathcal{P} := \{(k,i) : c'\sqrt{n} \leq k \leq 2c'\sqrt{n} \text{ and } 0 \leq i \leq c'\sqrt{n}\}.$$

For $(k,i) \in \mathcal{P}$, define $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ by taking the first $k-i$ elements from $\mathbf{S}$ and the first i elements from $\mathbf{T}$. By Property 2 and 3 of Lemma 3.6, the values of $e(U_0, \mathbf{Z}_{k,i}) := \sum_{\mathbf{z} \in \mathbf{Z}_{k,i}} d_{U_0}(\mathbf{z})$ are evenly spaced by $\Theta(\sqrt{n})$ as (k,i) vary lexicographically, which already forms $\Omega(n)$ distinct values of $e(U_0 \cup \mathbf{Z}_{k,i})$. It remains to fill the $\Theta(\sqrt{n})$ gaps between each $e(U_0 \cup \mathbf{Z}_{k,i})$ with $\mathbf{X}$ via the *double-exposure* technique while maintaining the gap struture of $\mathbf{Z}_{k,i}$. Specifically, we sample U from U_0 such that the values $e(U \cup \mathbf{Z}_{k,i} \cup \{\mathbf{x}\})$ are distinct as $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ and $\mathbf{x} \in \mathbf{X}$ vary. Define $U_{k,i} := U \cup \mathbf{Z}_{k,i}$ and $e_{k,i} := e(U_{k,i}) - e(U) = e(\mathbf{Z}_{k,0}) + e(\mathbf{Z}_{k,0}, U)$. We need to ensure the following:

1. *Degree distinctness in* $\mathbf{X}$. For most (k,i), the elements of $\mathbf{X}$ should have $\Omega(\sqrt{n})$ pairwise distinct degrees into $U_{k,i}$, all lying in an interval of length $O(\sqrt{n})$. This is the payoff of double-exposure.

2. *Concentration of the edge count.* The value $e_{k,0}$ should be $O(n)$ around its expectation, so that the $\Theta(n)$ spacings between $e_{k,0}$ and $e_{k+1,0}$ are perserved.

3. *Regular medium-scale spacing.* For each k, the values $e_{k,0}, e_{k,1}, \ldots, e_{k,c'\sqrt{n}}$ should span a range of $\Omega(n)$ and increase roughly uniformly, so that $\Theta(\sqrt{n})$ of them are pairwise separated by $\Omega(\sqrt{n})$.

Write $\Delta_{k,i} := e_{k,i} - e_{k,i-1}$ and we have the following lemma, whose first two properties correspond to the first two points above, and the last two properties combined ensures the regular medium-scale spacing.

Lemma 3.8. *Let G be a C-Ramsey graph, let $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$, U_0, c', d be obtained from Lemma 3.6. Let U be a random subset of U_0 where each vertex is present with probability $1/2$, and let $U_{k,i}, e_{k,i}, \Delta_{k,i}$ be defined as above. Then there exist positive constants $M, \beta, Q > 0$ depending on C, such that $Q \geq 3\beta$ and for each $c'\sqrt{n} \leq k \leq 2c'\sqrt{n}$ the following hold:*

1. *With probability ≥ 0.99, there is a set $\mathcal{I}_k$ of $(1 - \beta/(2M)) \cdot c'\sqrt{n}$ values with the following property: for each $i \in \mathcal{I}_k$, there exists $\mathbf{X}_{k,i} \subseteq \mathbf{X}$ of size $\Omega(\sqrt{n})$, such that the values $d_{U_{k,i}}(\mathbf{x})$ for $\mathbf{x} \in \mathbf{X}_{k,i}$ are distinct and all lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.*

2. *With probability ≥ 0.99,*
$$|e_{k,0} - \mathbb{E}[e_{k,0}]| \leq Qn.$$

3. *With probability $\geq 1/2$,*
$$e_{k,c'\sqrt{n}} - e_{k,0} \geq 3\beta n.$$

4. *With probability at least 0.99,*
$$\sum_{i\,:\,|\Delta_{k,i}| \geq M\sqrt{n}} |\Delta_{k,i}| \leq \beta n.$$

   That is, the unusually large increments contribute little to the total range $e_{k,c'\sqrt{n}} - e_{k,0}$.

Proof. We prove each property in the order of 3, 4, 1, 2, so that the constants β, M, Q are determined in their dependency order.

3. Note that $|e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})| \leq 2c'\sqrt{n}$, as $|\mathbf{Z}_{k,i}| = k \leq 2c'\sqrt{n}$. This and the construction of U yields
$$\begin{aligned}\mathbb{E}[e_{k,i} - e_{k,i-1}] &= \mathbb{E}[e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,i-1})] + [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})] \\ &\geq \frac{1}{2}[e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1})] - 2c'\sqrt{n}.\end{aligned}$$
   Property 3 of Lemma 3.6 gives $e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1}) \geq 8c'\sqrt{n}$, so
$$\mathbb{E}[e_{k,i} - e_{k,i-1}] \geq 2c'\sqrt{n},$$
   for all i. Put $\beta = 2(c')^2/3$ so the $e_{k,c'\sqrt{n}} - e_{k,0}$ has expectation $\geq 3\beta n$. Since $e(U, \mathbf{Z}_{k,i}) = \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot d_{\mathbf{Z}_{k,i}}(u)$,
$$\begin{aligned}e_{k,c'\sqrt{n}} - e_{k,0} &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + [e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,0})] \\ &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot (d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,0}}(u))\end{aligned}$$
   is of Littlewood–Offord type with probability $1/2$, which is symmetrically distributed around its expectation. The result now follows.

4. Note that

$$\Delta_{k,i} = \Big[e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})\Big] + \sum_{u\in U_0} \mathbb{1}_{u\in U} \cdot \Big(d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u)\Big)$$

is of Littlewood–Offord type and measures the edge count difference of swapping one vertex from $\mathbf{S}$ to $\mathbf{T}$, so by Lemma 3.6 $\mathbb{E}\Big[\Delta_{k,i}\Big] = O(\sqrt{n})$ and each $u \in U_0$ contributes at most $|d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u)| \le 2$ to $\Delta_{k,i}$. Thus by Hoeffding's inequality (Proposition 1.8),

$$\mathbb{P}[|\Delta_{k,i}| \ge t] \le 2\exp\left(\frac{2(t - \mathbb{E}[|\Delta_{k,i}|])^2}{4|U_0|}\right) = \exp(-\Omega(t^2/n)).$$

By the identity $\mathbb{E}[X] = \sum_{t\ge 1} \mathbb{P}(X \ge t)$ for nonnegative integer random variable X,

$$\begin{aligned}
\mathbb{E}\Big[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\ge M\sqrt{n}}\Big] &= \sum_{t=1}^{\infty} \mathbb{P}\Big[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\ge M\sqrt{n}} \ge t\Big] \\
&= M\sqrt{n} \cdot \mathbb{P}\Big[|\Delta_{k,i} \ge M\sqrt{n}|\Big] + \sum_{t=M\sqrt{n}}^{\infty} \mathbb{P}\Big[|\Delta_{k,i}| \ge t\Big] \\
&= M\sqrt{n} \cdot \exp\Big(-\Omega(M^2)\Big) + \sum_{t=M\sqrt{n}}^{\infty} \exp\Big(-\Omega(t^2/n)\Big) \\
&= \sqrt{n} \cdot \exp\Big(-\Omega(M^2)\Big),
\end{aligned}$$

uniformly over M. For large enough M, Markov's inequality now yields

$$\mathbb{P}\left[\sum_{i\,:\,|\Delta_{k,i}|\ge M\sqrt{n}} |\Delta_{k,i}| > \beta n\right] \le \frac{\sum_{i=1}^{|\mathbf{S}|} \mathbb{E}\Big[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\ge M\sqrt{n}}\Big]}{\beta n} \le \frac{e^{-\Omega(M^2)}}{\beta} < 0.99.$$

1. For $\mathbf{x}, \mathbf{y} \in \binom{\mathbf{X}}{2}$,

$$d_U(\mathbf{x}) - d_U(\mathbf{y}) = \sum_{u\in N_U(\mathbf{x})} \mathbb{1}_{u\in U} - \sum_{u\in N_U(\mathbf{y})} \mathbb{1}_{u\in U} = \sum_{u\in N_U(\mathbf{x})\triangle N_U(\mathbf{y})} a_u \mathbb{1}_{u\in U}$$

with $a_u \in \{1, -1\}$. Thus for each k, i, the random variable $d_{U_{k,i}}(\mathbf{x}) - d_{U_{k,i}}(\mathbf{y})$ is of $(|N_U(\mathbf{x})\triangle N_U(\mathbf{y})|, 1/2)$-Littlewood–Offord type, as it's a deterministic translation of $d_U(\mathbf{x}) - d_U(\mathbf{y})$. By Property 4 of Lemma 3.6 and Littlewood–Offord theorem (Proposition 1.10),

$$\mathbb{P}[d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})] = O(1/\sqrt{n}).$$

Let $H_{k,i}$ be the graph on $\mathbf{X}$ with edges $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{X}}{2}$ such that $d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})$, so $\mathbb{E}[e(H_{k,i})] = |\mathbf{X}|^2 \cdot O(1/\sqrt{n}) = O(\sqrt{n})$. By Markov's inequality, for large enough n,

$$\mathbb{P}[e(H_{k,i}) = O(\sqrt{n})] \ge 1 - \frac{\mathbb{E}[e(H_{k,i})]}{\omega(\sqrt{n})} \ge 1 - \frac{\beta}{400M},$$

in which case Turán's theorem (Proposition 1.9) yields an independent set $\mathbf{Y}_{k,i}$ of size $n/(2O(\sqrt{n})+\sqrt{n}) = 2\gamma\sqrt{n}$, for some constant $\gamma = \gamma(\beta, M) > 0$. In particular, $d_{U_{k,i}}(\mathbf{x})$ is distinct for $\mathbf{x} \in \mathbf{Y}_{k,i}$. The expected proportion of values of i for which this fails to occur is $\beta/400M$, and by Markov's inequality again, it fails only for $\beta/2M$ proportion of values of i with probability at least $1-(\beta/400M)/(\beta/2M) = 0.995$, in which case put $\mathcal{I}_k$ as the set of i that succeeds.

By Property 2 of Lemma 3.6, $\mathbb{E}[d_U(\mathbf{x})] = d/2 + O(\sqrt{n})$ for each $\mathbf{x} \in \mathbf{X}$. Since $d_U(\mathbf{x})$ follows a binomial distribution with parameters $O(n)$ and $1/2$, we have $\mathrm{Var}(d_U(\mathbf{x})) = O(n)$. By triangle inequality,

$$|d_U(\mathbf{x}) - d/2| \le |d_U(\mathbf{x}) - \mathbb{E}[d_U(\mathbf{x})]| + O(\sqrt{n}),$$

so for sufficiently large Q, Chebyshev's gives

$$\mathbb{P}\left[|d_U(\mathbf{x}) - d/2| > Q\sqrt{n}\right] \le \frac{\mathrm{Var}(d_U(\mathbf{x}))}{Q^2 n} = \frac{O(1)}{Q^2} < \frac{\gamma}{200}.$$

Let $\mathbf{Y}$ be the set of all $\mathbf{x} \in \mathbf{X}$ such that $|d_U(\mathbf{x}) - d/2| \le Q\sqrt{n}$. Then each $\mathbf{x} \in \mathbf{X}$ is not present in $\mathbf{Y}$ with probability $< \gamma/200$. By Markov's,

$$\mathbb{P}\left[|\mathbf{Y}| \ge (1-\gamma)|\mathbf{X}|\right] = 1 - \mathbb{P}\left[|\mathbf{X} \setminus \mathbf{Y}| \ge \gamma|\mathbf{X}|\right] \ge 1 - \frac{\gamma|\mathbf{X}|}{200\gamma|\mathbf{X}|} = 0.995.$$

Hence, with probability ≥ 0.995, there exists large enough Q and set $\mathbf{Y} \subseteq \mathbf{X}$ of size $\ge (1-\gamma)|\mathbf{X}|$ whose elements lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.

By union bound, both above events occur with probability ≥ 0.99, in which case take $\mathbf{X}_{k,i} = \mathbf{Y}_{k,i} \cap \mathbf{Y}$ for $i \in \mathcal{I}_k$. Since $|\mathbf{Y}_{k,i}| \ge 2\gamma\sqrt{n}$ and $|\mathbf{Y}| \ge (1-\gamma)|\mathbf{X}|$, we have $|\mathbf{X}_{k,i}| \ge \gamma\sqrt{n}$. This proves Property 1.

2. Since

$$e_{k,i} = e(\mathbf{Z}_{k,i}) + e(U, \mathbf{Z}_{k,i}) = e(\mathbf{Z}_{k,i}) + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot d_{\mathbf{Z}_{k,i}}(u),$$

$e_{k,i}$ is of $(O(n), 1/2)$-Littlewood–Offord type with coefficients $d_{\mathbf{Z}_{k,i}}(u) = O(\sqrt{n})$. Thus $\mathrm{Var}(e_{k,0}) = O(n^2)$ and by Chebyshev's,

$$\mathbb{P}\left[|e_{k,0} - \mathbb{E}[e_{k,0}]| \le Qn\right] \ge 1 - \frac{\mathrm{Var}(e_{k,0})}{Q^2 n^2} = 0.99,$$

for large enough Q. We end this proof by noting that enlarging Q does not fail Property 1.

□

3.4 Proof of the main lemma

We assemble Lemma 3.6 and Lemma 3.8 to prove the main lemma.

Apply Lemma 3.6 to obtain $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}, c', d$, and define $\mathcal{P}, U, \mathbf{Z}_{k,i}, U_{k,i}$ as in Section 3.3. Note that U is the only source of randomness. We prove the main lemma with W being the set of all vertices in $\mathbf{S}, \mathbf{T}, \mathbf{X}$.

Recall that k ranges over the interval between $c'\sqrt{n}$ and $2c'\sqrt{n}$. For each k, union bound shows all four properties of Lemma 3.8 hold with probability ≥ 0.4. Let $\mathcal{K}$ be the set of all such k. Then $\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right] \leq 0.6\, c'\sqrt{n}$, and Markov's inequality yields

$$\mathbb{P}\left[|\mathcal{K}| \geq 0.2\, c'\sqrt{n}\right] = 1 - \mathbb{P}\left[c'\sqrt{n} - |\mathcal{K}| \geq 0.8\, c'\sqrt{n}\right] \geq 1 - \frac{\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right]}{0.8\, c'\sqrt{n}} \geq 0.25.$$

We now show that $e(U)$ is close to m with high probability. Note that $|m - \mathbb{E}[e(U)]| = O(n^{3/2})$, as $\mathbb{E}[e(U)] = e(U_0)/4$ and $|e(U_0) - 4m| = O(n^{3/2})$ from Lemma 3.6. Recall that $\mathrm{Var}(e(U)) = O(n^3)$. Since $|e(U) - m| \leq |e(U) - \mathbb{E}[e(U)]| + |\mathbb{E}[e(U)] - m|$, Chebyshev's inequality yields

$$\mathbb{P}\left[|e(U) - m| = O(n^{3/2})\right] \geq \mathbb{P}\left[|e(U) - \mathbb{E}[e(U)]| = O(n^{3/2})\right] \geq 1 - \frac{\mathrm{Var}(e(U))}{\omega(n^3)} \geq 0.9,$$

for large enough n. Fix an outcome U satisfying both events above, which hold together with probability $\geq 0.25 + 0.9 - 1 = 0.15$. Let $Q, \beta, M, \mathcal{I}_k$ be given by Lemma 3.8.

For each k,

$$\mathbb{E}[e_{k,0}] - \mathbb{E}[e_{k-1,0}] = \frac{1}{2}[e(\mathbf{Z}_{k,0}, U_0) - e(\mathbf{Z}_{k-1,0}, U_0)] + O(\sqrt{n}) = \Theta(n),$$

as $\mathbb{E}[d_U(\mathbf{x})] = d_{U_0}(\mathbf{x})/2 = \Theta(n)$ for $\mathbf{x} \in \mathbf{S}$. We find a subset of $\mathcal{K}$ whose realized values $e_{k,0}$ are well-separated by $\Theta(n)$. Specifically, let $\mathcal{K}' \subseteq \mathcal{K}$ contain every qth element of $\mathcal{K}$, for sufficiently large q such that the values of $\mathbb{E}[e_{k,0}]$ are separated by $\geq 4Qn$ for $k \in \mathcal{K}'$. Then, $|\mathcal{K}'| = \Theta(\sqrt{n})$ and Property 2 of Lemma 3.8 ensures the values of $e_{k,0}$ are separated by $\geq 2Qn$ for $k \in \mathcal{K}'$.

We now partition the linear gaps between the values of $e_{k,0} : k \in \mathcal{K}'$ into square-root sized gaps. Let $k \in \mathcal{K}'$. By Property 3 of Lemma 3.8, $e_{k,c'\sqrt{n}} - e_{k,0} \geq 3\beta n$. Consider the range of integers between $e_{k,0}$ and $e_{k,0} + 3\beta n$. Within this range, we take $2\beta\sqrt{n}/M$ intervals of length $M\sqrt{n}$, each separated by a distance of $M\sqrt{n}/2 = \Omega(\sqrt{n})$. By Property 4 of Lemma 3.8, $\leq \beta\sqrt{n}/M$ of these intervals contain no value of $e_{k,i}$. Consider the $\beta\sqrt{n}/M$ intervals that do contain a value of $e_{k,i}$ and pick a representative $e_{k,i}$ from each. Let $\mathcal{I}'_k$ be the set of indices i corresponding to these representatives, and let $\mathcal{I}''_k = \mathcal{I}_k \cap \mathcal{I}'_k$. Then

$$|\mathcal{I}''_k| = |\mathcal{I}'_k| + |\mathcal{I}'_k| - |\mathcal{I}'_k \cup \mathcal{I}_k| \geq \beta\sqrt{n}/M - \beta c'\sqrt{n}/2M = \Omega(\sqrt{n})$$

and $e_{k,i}$ are separated by $\Omega(\sqrt{n})$ for $i \in \mathcal{I}''_k$. Recall that $Q \geq 3\beta$, so by construction $|e_{k,i} - e_{k,0}| \leq 3\beta n \leq Qn$ for $i \in \mathcal{I}''_k$. In particular, amongst choices of $k \in \mathcal{K}'$ and $\mathcal{I}''_k$, we already have $\Omega(n)$ distinct values of $e(U_{k,i})$ that are separated by $\Omega(\sqrt{n})$. It remains to fill the square-root sized gaps between these values.

In increasing order of $e(U_{k,i})$, let $\mathcal{P}' \subseteq \{(k,i) : k \in \mathcal{K}', i \in \mathcal{I}''_k\}$ be the corresponding index set of every qth of these values. We enlarge q so that $e(U_{k,i})$ from $\mathcal{P}'$ are separated by $\geq 2Qn$. For each $(k,i) \in \mathcal{P}'$ we have $i \in \mathcal{I}_k$, so Property 1 of Lemma 3.8 furnishes a set $\mathbf{X}_{k,i}$ such that each $d_{U_{k,i}}(\mathbf{x})$ is unique for $\mathbf{x} \in \mathbf{X}_{k,i}$ while lying in a fixed interval of length $2Qn$ centered at $d/2$, where d is independent of k and i. For each $(k,i) \in \mathcal{P}'$, the set

$$\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}$$

lie in an interval of length $2Qn$ centered at $e(U_{k,i}) + d/2$, so these sets do not overlap among different $(k,i) \in \mathcal{P}'$.

Therefore,

$$|\{e(U \cup Z) : Z \subseteq W\}| \geq |\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : (k,i) \in \mathcal{P}',\, \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}| = \Omega(n^{3/2}).$$

□

4 The Erdős–McKay Conjecture

Both proofs in Chapters 2 and 3 rely on the same anticoncentration principle that, when the symmetric difference $|N(x) \triangle N(y)|$ is large, the degree difference $\deg_U(x) - \deg_U(y)$ is a sum of many weakly dependent bounded random variables, and its point probabilities are $O(1/\sqrt{n})$. The Erdős–McKay conjecture asks for a qualitatively stronger result that asserts $\Phi(G)$ to contain every integer in a quadratic-length interval, not merely that it is large. The conjecture was resolved by Kwan, Sah, Sauermann, and Sawhney [9], and this chapter is dedicated to outlining their proof.

Theorem 4.1 (Erdős–McKay conjecture [9])**.** *Fix $C > 0$. Let G be a C-Ramsey graph on n vertices for sufficiently large n. Then there exists $\delta = \delta(C) > 0$ such that*

$$\Phi(G) \supseteq \{1, 2, \ldots, \delta e(G)\}.$$

The approach to this theorem fundamentally shifts from bounding degree-collision probabilities for pairs of vertices to directly controlling the point probabilities of $e(U)$ itself. This requires a substantially richer set of tools, but the underlying principle remains: when a random variable is approximately normal with large standard deviation, its point probabilities are small.

4.1 Point probabilities of $e(U)$

Assume that $V(G) = [n]$. Let U be a uniformly random vertex subset. By the Erdős–Szemerédi bound (Corollary 1.4), the random variable $e(G[U])$ has mean $e(G)/2 = \Theta(n^2)$ and standard deviation $\sigma(e(U)) = \Theta(n^{3/2})$.

In Chapters 2 and 3, the anticoncentration analysis targeted the degree difference $\deg_U(x) - \deg_U(y)$ with standard deviation $\Theta(\sqrt{n})$, and the control of $e(U)$ was achieved indirectly through the construction of $\mathbf{S}, \mathbf{T}, \mathbf{X}$. In [9], we show directly that the point probabilities of $e(U)$ are all of order $\Theta(1/\sigma(e(U))) = \Theta(n^{-3/2})$ within the bulk of the distribution.

Theorem 4.2 (Kwan–Sah–Sauermann–Sawhney [9])**.** *Fix constant $C > 0$. Let G be a C-Ramsey graph on n vertices and let U be a $(1/2)$-random subset of $V(G)$. Then*

$$\sup_{x \in \mathbb{Z}} \mathbb{P}[e(U) = x] = O(n^{-3/2}),$$

and for every fixed $A > 0$,

$$\inf_{\substack{x \in \mathbb{Z} \\ |x - e(U)/4| \le An^{3/2}}} \mathbb{P}[e(U) = x] = \Omega(n^{-3/2}).$$

This theorem is essentially an approximation of the local limit theorem for $e(U)$ up to constant factors. A true local CLT would give $\mathbb{P}[e(U) = x] \sim \varphi((x - \mu)/\sigma)/\sigma$, where φ is the standard Gaussian density function.

The Erdős–McKay conjecture follows directly from the lower bound of Theorem 4.2 combined with an earlier result on small values of x by Alon, Krivelevich, and Sudakov.

Proof of Theorem 4.1. By Theorem 3.1, there exists $\alpha = \alpha(C) > 0$ such that $\Phi(G) \subseteq \{0, 1, \ldots, n^\alpha\}$. Fix integer x between n^α and $e(G)/4$. Let m be the smallest integer such that $e(\{1, \ldots, m\}) \geq 4x$. Put $G' = G[\{1, \ldots, m\}]$ and note that

$$e(G') \geq 4x \geq e(\{1, \ldots, m-1\}) \geq e(G') - m,$$

so $|x - e(G')/4| \leq m/4 \leq m^{3/2}$. Since $m^2 \geq e(G') \geq 4x \geq n^\alpha$, we have $m \geq n^{\alpha/2}$, so G' is $(2C/\alpha)$-Ramsey. Applying Theorem 4.2 to G' with $A = 1$ gives

$$\mathbb{P}[e(G'[U]) = x] = \Omega(m^{-3/2}) = \Omega(n^{3\alpha/4}) > 0.$$

Therefore, there exists $U \subseteq V(G') \subseteq V(G)$ such that $e(G[U)] = e(G'[U]) = x$, for sufficiently large n. This completes the theorem. □

We will outline the proof of Theorem 4.2 in the remainder of this chapter.

4.2 Proof overview

The standard way to approach local limit theorems is via Fourier transforms. Define the *graph polynomial* as

$$f(\xi_1, \ldots, \xi_n) := \sum_{ij \in E(G)} \xi_i \xi_j.$$

For any $U \subseteq V(G)$, setting $\xi_v = \mathbb{1}_{v \in U}$ gives $f(\vec{\xi}^{(U)}) = e(U)$. The Erdős–McKay conjecture is thus equivalent to asking whether the range of this quadratic polynomial on $\{0, 1\}^n$ covers every integer in $\{0, \ldots, \delta n^2\}$.

For a random variable X, the *characteristic function* of X is defined as

$$\varphi_X(\tau) := \mathbb{E}\left[e^{i\tau X}\right] = \int_{-\infty}^{\infty} e^{i\tau X} \cdot \mathbb{P}[X = x]\, dx, \qquad \tau \in \mathbb{R},$$

and note that $|\varphi_X(\tau)| \leq 1$ for all τ. Applying the Fourier inversion formula yields

$$\mathbb{P}[X = x] = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i\tau x} \varphi_X(\tau)\, d\tau \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\varphi_X(\tau)|\, d\tau, \tag{2}$$

for integers-valued X. In particular, any upper bound on the integrated $|\varphi_X(\tau)|$ produces an upper bound on the point probability. For the matching lower bound, we compare φ_X with the characteristic function $\varphi_Z(\tau) = e^{i\tau\mu - \sigma^2\tau^2/2}$ of a Gaussian $Z \sim \mathcal{N}(\mu, \sigma^2)$, where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$. The Gaussian density at x is

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\tfrac{(x-\mu)^2}{2\sigma^2}\right),$$

which is of order $\Theta(1/\sigma)$ for any x within $O(\sigma)$ of the mean. Hence, if $|\varphi_X(\tau)-\varphi_Z(\tau)|$ is small, then $\mathbb{P}[X = x]$ inherits the same order of magnitude $\Theta(1/\sigma)$ through (2). If a local central limit theorem (CLT) held for $X = e(U)$, the proof would amount to verifying suitable estimates on $|\varphi_X(\tau)|$ on $[-\pi, \pi]$ and we are done.

Unfortunately, an actual local CLT does not hold for $e(U)$ in general. The distribution of $e(U)$ can fail to be smooth at the unit scale despite the global shape being approximately Gaussian, which results in spikes in the characteristic function and prevents the integrand in (2) from being uniformly small.

The proof of Theorem 4.2 overcomes this obstacle in two stages. The first stage splits into two cases according to whether the degree sequence of G is additively structured or not. The *unstructured case* can be handled by a standard Fourier-analytic argument without further intervention. The *additively structured case* is where $|\varphi_X(\tau)|$ splikes, which we will dedicate most of the effort to handle. A Fourier-analytic argument on these two cases yield the interval estimate

$$\mathbb{P}\left[|X - x| \leq B\right] = \Theta(n^{-3/2}), \tag{3}$$

for some constant $B = B(C)$. The second stage upgrades it to a pointwise estimate by showing that the probability mass cannot be too unevenly distributed within the interval via a *switching argument*.

4.3 Degree sequence dichotomy

We first decompose $e(U)$ into a sum of a linear and quadratic part. Set $\xi_v = \mathbb{1}_{v\in U}$ and write $x_v = 2\xi_v - 1 \in \{-1, +1\}$, so that $\xi_v = (1 + x_v)/2$. Substituting into the $e(U) = f(\vec{\xi}) = \sum_{ij\in E(G)} \xi_i\xi_j$ yields

$$X = \sum_{ij\in E(G)} \frac{(1 + x_i)(1 + x_j)}{4} = \underbrace{\frac{e(G)}{4}}_{\mathbb{E}[X]} + \underbrace{\frac{1}{4}\sum_{v\in V(G)} \deg(v)x_v}_{L} + \underbrace{\frac{1}{4}\sum_{uv\in E(G)} x_u x_v}_{Q}, \tag{4}$$

and we denote the linear term as L and the quadratic term as Q. Since $\mathbb{E}[x_v] = 0$ and $\mathbb{E}[x_u x_v] = 0$ for $u \neq v$, we have that $\mathbb{E}[L] = \mathbb{E}[Q] = 0$, the covariance of L and Q is zero, and the constant term $e(G)/4 = \mathbb{E}[X]$. It can be shown via the Erdős–Szemerédi bound and Jensen's inequality that $\mathrm{Var}(L) = \Theta(n^3), \mathrm{Var}(Q) = \Theta(n^2)$. Therefore, the linear part L dominates the total variance, so we expect L to drive the global behaviour of X. Write $d_v \coloneqq \deg(v)/4$ so that $L = \sum_v d_v x_v$. Since the variables x_v are independent,

$$\varphi_L(\tau) = \mathbb{E}\left[\exp\left(i\tau \sum_v d_v x_v\right)\right] = \prod_{v\in V(G)} \mathbb{E}\left[\exp(i\tau d_v x_v)\right] = \prod_{v\in V(G)} \cos(\tau d_v). \tag{5}$$

The decomposition (4) thus allows us to bound X by bounding the characteristic function of L which factors well.

For $y \in \mathbb{R}$, denote $\|y\| := \operatorname{dist}(y, \mathbb{Z})$ as the distance from y to the nearest integer, and note that $0 \le \|y\| \le 1/2$. By Jordan's inequality and $1 - t \le \exp(-t)$ for $t \ge 0$,

$$|\cos(\pi y)| = \cos(\pi\|y\|) = \sqrt{1-\sin^2(\pi\|y\|)} \le \sqrt{1-4\|y\|^2} \le 1-2\|y\|^2 \le \exp(-2\|y\|^2),$$

for $y \in \mathbb{R}$, so applying to (5) gives

$$|\varphi_L(\tau)| \le \exp\left(-2 \sum_{v \in V(G)} \left\| \frac{\tau d_v}{\pi} \right\|^2\right). \tag{6}$$

In particular, the right-hand side decays exponentially in n unless many of the quantities $\tau d_v/\pi$ are simultaneously close to integers. We treat these two cases separately. Throughout what follows, let $\nu = \nu(C) > 0$ be a small constant and take the relevant frequency window to be $0 < |\tau| \le \nu$.

The unstructured case. In the case that

$$\sum_{v \in V(G)} \left\| \frac{\tau d_v}{\pi} \right\|^2 = \omega(\log n)$$

for all τ with $0 < |\tau| \le \nu$, the bound (6) gives the superpolynomial decay $|\varphi_L(\tau)| \le n^{-\omega(1)}$ on the window. In this case, Esseen's inequality applied to the difference $\varphi_X - \varphi_Z$, where $\varphi_Z(\tau) = \exp(i\mu\tau - \sigma^2\tau^2/2)$ is the Gaussian characteristic function matching the mean and variance of X, then delivers the interval estimate (3). This case admits a local central limit theorem by the standard Fourier-analytic argument.

The structured case. Here we consider the opposite case that

$$\sum_{v \in V(G)} \left\| \frac{\tau d_v}{\pi} \right\|^2 = O(\log n),$$

for some τ with $0 < |\tau| \le \nu$. This says that for this one frequency, many of the numbers $\tau d_v/\pi$ are simultaneously close to integers. Lemma 4.12 of [9] uses this condition to show that for small enough $\gamma > 0$, the vertex set admits a partition

$$V(G) = R \sqcup (B_1 \sqcup \cdots \sqcup B_m),$$

with $|R| \le n^{1-\gamma}$, $|B_j| = \lceil n^{1-2\gamma} \rceil$ for $j \in [m]$, and real numbers $\kappa_1, \ldots, \kappa_m \ge 0$ such that

$$|d_v - \kappa_j| \le n^{1/2+4\gamma},$$

for $j \in [m]$ and $v \in B_j$. In other words, the vertex set splits into $m = \Theta(n^{2\gamma})$ many buckets $B_1, \ldots, B_m$ each with degrees clustered around κ_j, together with an sublinear sized exceptional set R. This allows us to write L as

$$L = \sum_{j=1}^{m} \sum_{v \in B_j} d_v x_v + \sum_{v \in R} d_v x_v = \underbrace{\sum_{j=1}^{m} \kappa_j \sum_{v \in B_j} x_v}_{L_{main}} + \underbrace{\sum_{j=1}^{m} \sum_{v \in B_j} (d_v - \kappa_j) x_v}_{L_{err}} + \underbrace{\sum_{v \in R} d_v x_v}_{L_R}.$$

Then conditioning on

$$\vec{\Delta} := (|U \cap B_1|, \ldots, |U \cap B_m|, (x_v)_{v \in R})$$

makes L_{main} and L_R deterministic and reduces the conditional variance of L to $\mathrm{Var}(L_{err} \mid \vec{\Delta}) \leq n^{2+8\gamma}$. The effect of conditioning on $\vec{\Delta}$ can be thought of as conditioning on the approximate value of L which governs the large-scale behaviour of X, allowing us to "zoom in" to see the smaller-scale behaviour of the distribution. In particular, $U \setminus R$ is conditionally uniform over the product of Boolean slices

$$\binom{B_1}{\Delta_1} \times \cdots \times \binom{B_m}{\Delta_m}.$$

This reduces the interval estimate (3) to evaluating the small-ball probability of $L_{err} + Q$ on this product of Boolean slices.

It is tempting at this point to hope that, after conditioning on $\vec{\Delta}$, $L_{\mathrm{err}} + Q$ satisfies a local CLT of its own. This hope is too optimistic. Consider G to be the disjoint union of two independent copies of $G(n/2, 1/2)$. With high probability G is a Ramsey graph, but conditioning on $|U| = n/2$ yields a normalized limiting distribution of the form $\alpha Z^2 + \beta Z_2$ for independent Gaussians Z_1, Z_2. Therefore, the asymptotic conditional distribution of X need not be Gaussian after the bucket conditioning, though we shall see that it can always be evaluated at independent Gaussians.

4.4 The structured case

The target now is to show

$$\mathbb{P}\left[|X - x| \leq B \mid \vec{\Delta}\right] = O(n^{-1}), \tag{7}$$

which, after averaging over $\vec{\Delta}$, yields the interval estimate $\Theta(n^{-3/2})$ in (3). The averaging contributes the remaining factor $n^{-1/2}$, reflecting the fact that only a $\Theta(n^{-1/2})$ fraction of $\vec{\Delta}$ values place $\mathbb{E}[X \mid \vec{\Delta}]$ within $O(n)$ of x. As shown in the previous section, the conditional distribution of X need not be approximately Gaussian, so the standard Fourier-analytic approach does not close on its own. [9] overcomes this by combining the following three ingredients.

Ingredient 1. The Gaussian invariance principle We first replace the product of Boolean slices by a Gaussian space. The invariance principle of Mossel, O'Donnell and Oleszkiewicz says that a low-degree polynomial of independent bounded random variables has a distribution close to that of the same polynomial evaluated at independent standard Gaussians. Applying this to $L_{err} + Q$ produces a quadratic polynomial

$$f(\vec{Z}) = \vec{Z}^\top F \, \vec{Z} + \vec{f} \cdot \vec{Z} + f_0, \qquad \vec{Z} = (Z_1, \ldots, Z_n) \sim \mathcal{N}(0, I_n), \tag{8}$$

with a symmetric matrix $F \in \mathbb{R}^{n\times n}$ and $\vec{f} \in \mathbb{R}^n$ derived from the adjacency matrix of G and the bucket partition, such that the characteristic functions of $L_{err} + Q$ (given $\vec{\Delta}$) and of $f(\vec{Z})$ are close – see [9, Lemma 11.1]. The role of this step is to convert a discrete, dependent domain into a continuous, independent one, allowing us to apply the Carbery–Wright theorem.

Ingredient 2. The sharpened Carbery–Wright theorem Given the $f(\vec{Z})$ from ingredient 1, we now bound $\sup_x \mathbb{P}[|f(\vec{Z}) - x| \le 1]$. In our application $\sigma(f(\vec{Z})) = \Theta(n)$, since the invariance principle preserves the variance and (4) gave $\mathrm{Var}(Q) = \Theta(n^2)$. The target $O(n^{-1})$ in (7) is therefore asking for a bound of the form $1/\sigma(f(\vec{Z}))$.

The classical result in this direction is the Carbery–Wright theorem, which states that for quadratic polynomial f in independent standard Gaussians and any $\varepsilon > 0$,

$$\sup_{x\in\mathbb{R}} \mathbb{P}\left[|f(\vec{Z}) - x| \le \varepsilon\right] = O\left(\sqrt{\varepsilon/\sigma(f(\vec{Z}))}\right), \tag{9}$$

which is short of our target $O(1/\sigma(f(\vec{Z})))$. The square-root exponent in (9) cannot be improved in general, but it turns out that forcing the associated matrix F to have effective rank at least three yields the desired bound.

Theorem 4.3 (Sharpened Carbery–Wright, [9, Theorem 1.6])**.** *Let $f(\vec{Z}) = \vec{Z}^\top F\vec{Z} + \vec{f}\cdot\vec{Z} + f_0$ be a real quadratic polynomial in independent standard Gaussians, with F a nonzero symmetric matrix. Suppose that for some $\eta > 0$,*

$$\min_{\substack{\widetilde{F}\in\mathbb{R}^{n\times n}\\ \mathrm{rank}(\widetilde{F})\le 2}} \|F - \widetilde{F}\|_F^2 \ge \eta\|F\|_F^2. \tag{10}$$

Then for $\varepsilon > 0$,

$$\sup_{x\in\mathbb{R}} \mathbb{P}\left[|f(\vec{Z}) - x| \le \varepsilon\right] = O\left(\varepsilon/\sigma(f(\vec{Z}))\right).$$

The hypothesis (10) says that F is a fixed Frobenius-norm proportion away from every matrix of rank ≤ 2. Thus, it remains to show (10) for the matrix F arising from the bucket structure, and this is where the Ramsey hypothesis enters.

Ingredient 3. The robust rank lemma

Lemma 4.4 (Robust rank, [9, Lemma 10.1])**.** *Fix $C > 0$, $0 < \delta < 1$ and $r \in \mathbb{N}$. Let G be a C-Ramsey graph on n vertices with adjacency matrix A, and let $V(G) = I_1 \cup \cdots \cup I_m$ be a partition into equal parts with $n^{\delta/2} \le m \le 2n^\delta$. Then for every matrix $B \in \mathbb{R}^{n\times n}$ with $\mathrm{rank}(B[I_j \times I_k]) \le r$ for all $j, k \in [m]$,*

$$\|A - B\|_F^2 \;=\; \Omega_{C,r,\delta}(n^2).$$

The lemma says that the adjacency matrix of a Ramsey graph is at Frobenius distance $\Omega(n)$ from every matrix whose restriction to each bucket block is low rank.

To sum it up, applying ingredients 1, 3, and 2 in sequence yields the interval estimate in (7).

4.5 Switching for pointwise control

The first stage delivers the interval estimate

$$\mathbb{P}[|X - x| \le B] = \Theta(n^{-3/2}),$$

but leaves open the worst case where all of this mass sits on a single integer $x' \neq x$ inside the window $[x - B, x + B]$. We rule this out via a *switching argument* between different possibilities of the set U, and this upgrade the interval estimate to a pointwise bound. The interval estimate (3) already implies an upper bound on $\mathbb{P}[X = x]$, so it remains to show a lower bound.

For $y \in U$ and $z \notin U$, the *switch* (y, z) replaces U by $U' = (U \setminus \{y\}) \cup \{z\}$. This operation shifts X by

$$s(y, z; U) \coloneqq e(G[U']) - e(G[U]) = \deg_{U\setminus\{y\}}(z) - \deg_U(y).$$

Fix a set $T \subseteq V(G)^2$ of candidate ordered pairs, to be specified later, and let

$$Y_\ell \coloneqq \#\Big\{(y, z) \in T : y \in U,\ z \notin U,\ s(y, z; U) = \ell\Big\}$$

count candidate switches at U that shift X by exactly ℓ. Notice that each candidate switch with $y \in U, z \in U$, and $s(y, z; U) = \ell$ pairs U with the set $U' = (U \setminus \{y\}) \cup \{z\}$, and the switch (z, y) from U' recovers U and shifts X by $-\ell$. In particular, pairing $(U, (y, z))$ with $(U', (z, y))$ in this way is a bijection. Since the uniform measure on $\{0, 1\}^n$ assigns the same probability 2^{-n} to every U, the bijection gives the symmetry identity

$$\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{X=x}\Big] = \mathbb{E}\Big[Y_{-\ell} \cdot \mathbb{1}_{X=x+\ell}\Big], \tag{11}$$

for any integer x and ℓ. By Cauchy–Schwarz inequality,

$$\Big(\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{X=x}\Big]\Big)^2 \le \mathbb{E}\Big[(Y_\ell^2 \cdot \mathbb{1}_{X=x})^2\Big] \cdot \mathbb{E}\Big[(\mathbb{1}_{X=x})^2\Big].$$

Rearranging and applying (11) yields

$$\mathbb{P}[X = x] \ge \frac{\Big(\mathbb{E}\Big[Y_{-\ell} \cdot \mathbb{1}_{X=x+\ell}\Big]\Big)^2}{\mathbb{E}\Big[Y_\ell^2 \cdot \mathbb{1}_{X=x}\Big]}. \tag{12}$$

The goal now is to find ℓ and the candidate set T so that

$$\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{|X-x|\le B}\Big] \gtrsim \frac{|T|}{\sqrt{n}} \cdot n^{-3/2}, \tag{13}$$

$$\mathbb{E}\Big[Y_\ell^2 \cdot \mathbb{1}_{|X-x|\le B}\Big] \lesssim \left(\frac{|T|}{\sqrt{n}}\right)^2 \cdot n^{-3/2}. \tag{14}$$

The two factors have concrete meanings. The factor $n^{-3/2}$ is $\mathbb{P}[|X - x| \le B]$, supplied by the interval estimate (3). The factor $|T|/\sqrt{n}$ represents $\mathbb{E}[Y_\ell]$, which we now

compute. For any pair $(y,z) \in T$, $\mathbb{P}[y \in U, z \notin U] = 1/4$. Conditional on this event, setting $R_w = 2\mathbb{1}_{w \in U} - 1$ and $a_w = 2\mathbb{1}_{w \in N(z) \setminus N(y)} - 1$,

$$\begin{aligned} s(y,z;U) &= \deg_{U \setminus \{y\}}(z) - \deg_U(y) \\ &= \sum_{w \in N(z),\, w \neq y} \mathbb{1}_{w \in U} - \sum_{w \in N(y)} \mathbb{1}_{w \in U} \\ &= \frac{\deg(z) - \deg(y)}{2} + \frac{1}{2} \sum_{w \in N(y) \triangle N(z)} a_w R_w \end{aligned}$$

is a sum of $|N(y) \triangle N(z)|$ independent $\pm$Bernoulli$(1/2)$ contributions plus the deterministic shift $(\deg(z) - \deg(y))/2$. Therefore, if we can find T such that $\deg(z) - \deg(y) = \Theta(\sqrt{n})$ and $|N(y) \triangle N(z)| = \Theta(n)$ for all $(y,z) \in T$, then a local CLT gives

$$\mathbb{P}[s(y,z;U) = \ell \mid y \in U, z \notin U] = \Theta(1/\sqrt{n})$$

for all ℓ in an $O(\sqrt{n})$ window around the mean, which yields

$$\mathbb{E}[Y_\ell] = \frac{1}{4} \sum_{(y,z) \in T} \mathbb{P}[s(y,z;U) = \ell \mid y \in U, z \notin U] = \Theta(|T|/\sqrt{n}).$$

It remains to construct such a set T, which is produced by applying richness twice inside a linear-sized (ε, δ)-rich subgraph S_0, together with a pigeonhole partition of the degree sequence into $\deg(z) - \deg(y) = \Theta(\sqrt{n})$ – see [9, Lemma 13.9] for details. The second-moment bound (14) is more involved and follows a *dependent random choice* technique – see [9, Lemmas 13.1, 13.4, 13.9] for details.

5 Concluding Remarks

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A Appendices